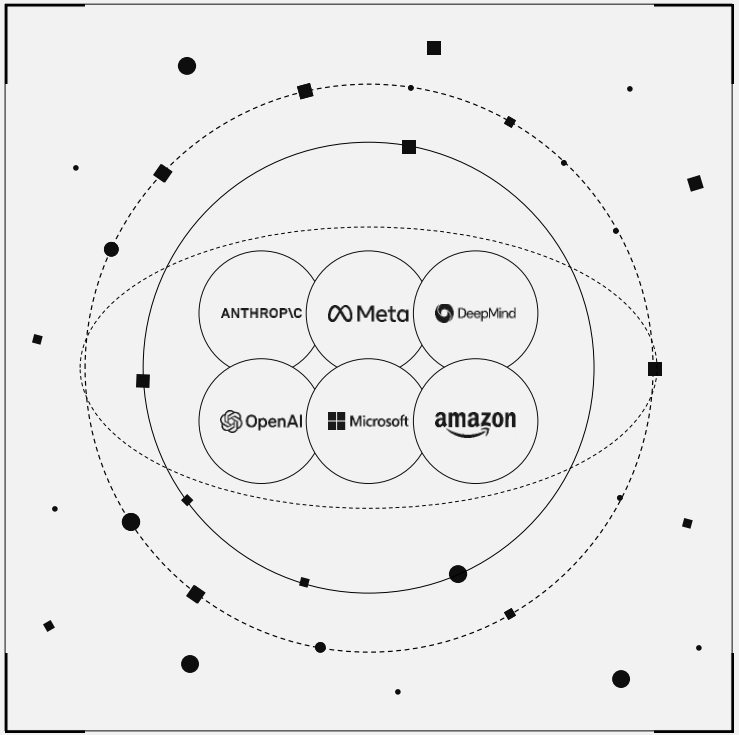


Optimizing AI Market Structure

AI progress is bottlenecked by centralization

[The current AI landscape is centralized,
resource-heavy, dominated by a few firms.]

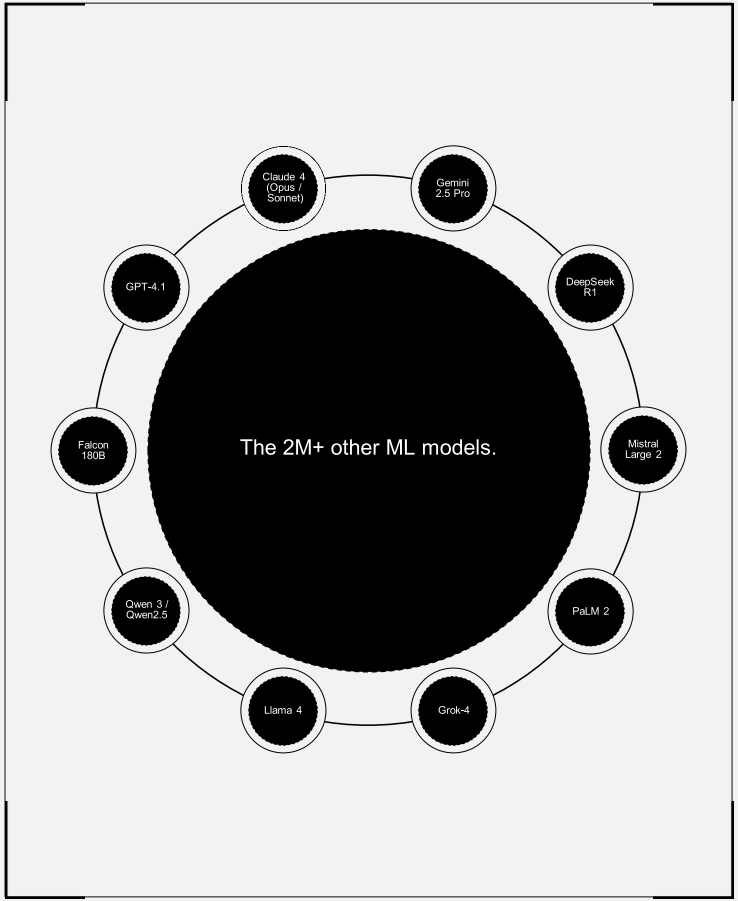


Presenting an untapped market
of ML models

// Total model supply is orders of magnitude greater than just that of centralized players offerings.

// Bigger brands ≠ better models. Modern open models approach or match top-tier performance for many tasks.

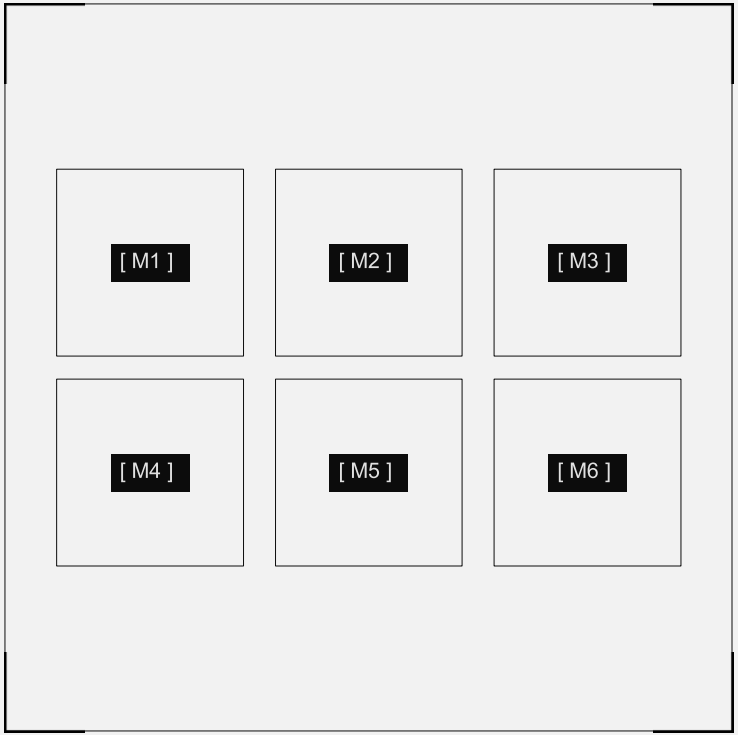
// Fragmented models suffer from inefficient coordination. The barrier is the system, not the availability.



Siloed Machine Intelligence

// The coordination of resources needed to build machine intelligence, coupling data, algorithms, and computational power, naturally favors industry monoliths.

// Building the best possible form of machine intelligence requires us to maximize the number of connections across a network in which diverse data sets and algorithms can be coupled freely, so that the most relevant insights can be obtained.

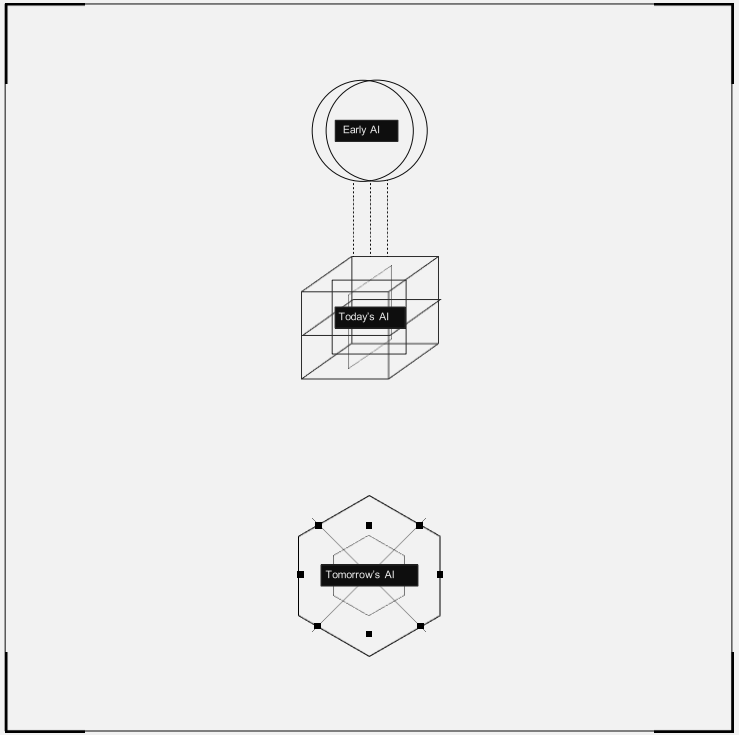


Market Structure Enabled
by Networks

// Shift from single entities → distributed networks.

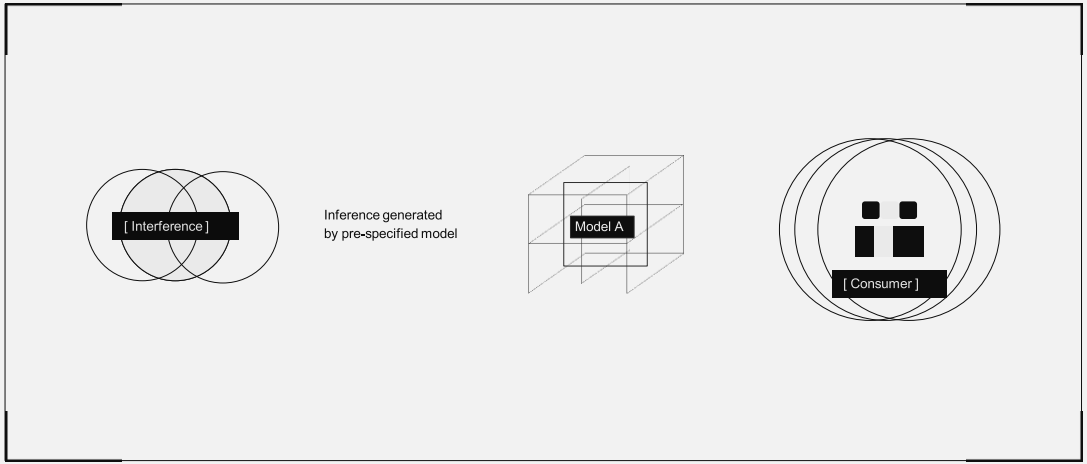
// Coordination of diverse models, data, and compute across participants.

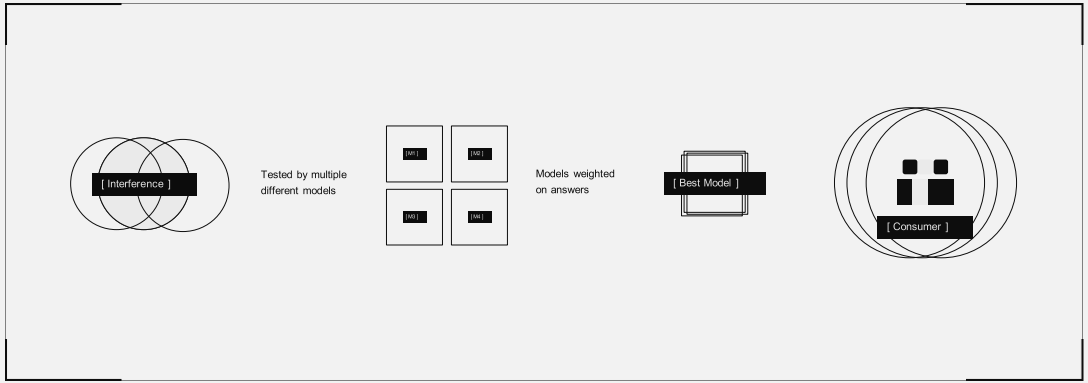
// More resilient, adaptive, and context-aware.



Model-Centric vs. Objective-Centric AI

[We face a big challenge: innovation is gated by who owns the infrastructure.]





Model-Centric

// Consumer specifies a specific model to interact with.

// They are asserting that this is the best model in every context in which they plan to use it.

Objective-Centric

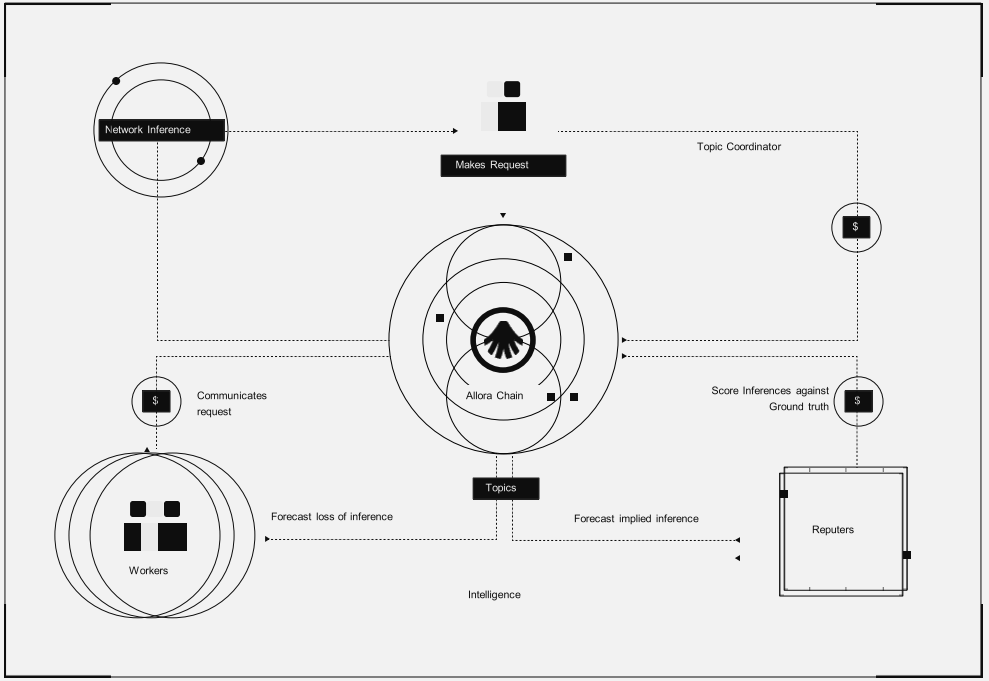
// Consumer specifies an ML objective instead of a specific model.

// An underlying set of models contribute to the collective optimization of the defined objective.

How to Produce Better AI

1. As requests come into the system, inference workers respond to those requests.
2. Inference workers share their responses with the topic coordinator and the forecasting workers.
3. Forecasting workers produce forecasted losses for the inference workers as requests are received.
4. Over time, forecasting workers learn the context in which different inference workers perform well or not and weight them accordingly.

// Additionally, because inference workers are only sharing outputs, the network can support both open and closed source models.



Context-Awareness

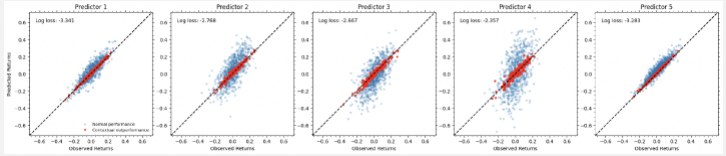
// These charts compare the performance of the network-level (aggregated) output vs individual model outputs.

// Inference workers are individual workers/ML models in a topics.

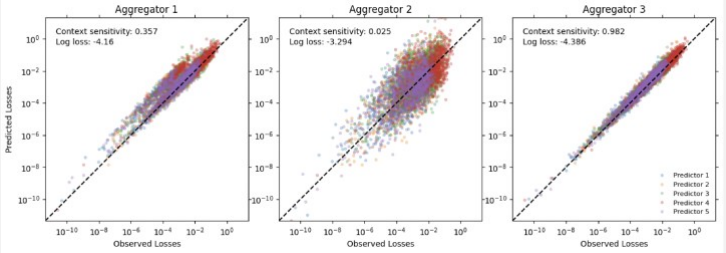
// Forecasters produce context-aware forecasted losses of inference workers' model outputs.

// As shown, the aggregated, network-level model output always outperforms the individual workers.

Figure 1: Shows the performance of inference workers (labeled predictors) on the network. As shown, Predictor 1 is the best with a log loss of -3.337.

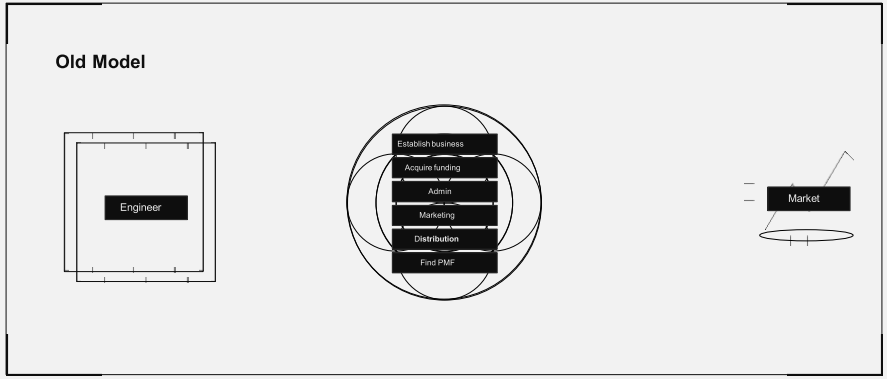


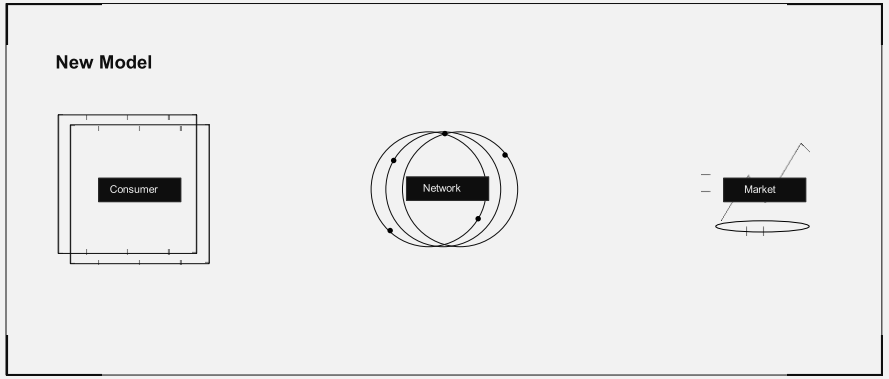
Shows the performance of network forecasts. These are context-aware forecasts of different individual workers' outputs. As shown, even the worst forecaster outperforms the best individual worker/predictor. The best forecaster achieves a log loss of -4.393, which is 3.8 times more accurate on average than the best individual worker.



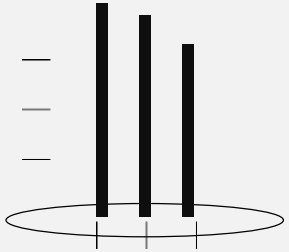
What This Means for ML Engineers

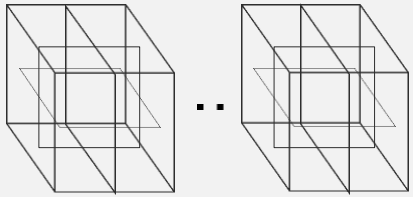
// Decentralized AI networks create a hyper-efficient market where ML engineers can directly capture value from their contributions.

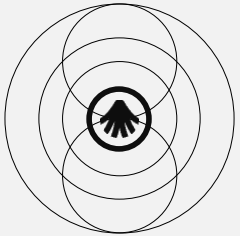




Who we are







2020-2021

// Market leader and early mover in decentralized
crowdsourcing research and mechanism design.

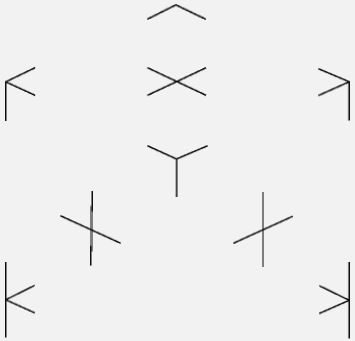
2021-2023

// Market leader and early mover in building critical
AI-powered crypto infrastructure.

2023 - Present

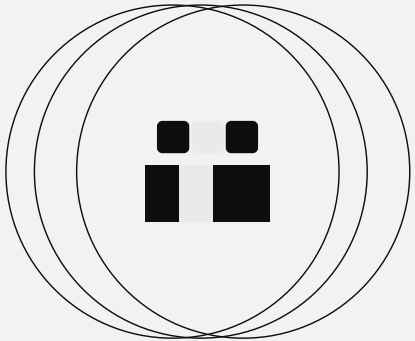
// Combining our work in breakthrough crowdsourcing research and early decentralized AI tooling to build a self-improving decentralized AI network: Allora.

The Network in Numbers



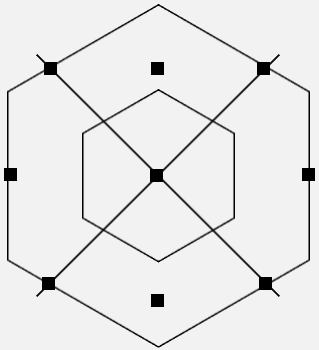
692M+

Inferences submitted to date



288K+

Workers



55+

Topics

RoboNet: Polymarket Trading Agent

// Overview: Allora provided real-time inferences that powered Pauly, an autonomous trading agent deployed by RoboNet to trade on Polymarket ahead of the 2024 U.S. Presidential Election.

// Allora's Role: Allora delivered high-precision predictions on Trump's election odds, enabling Pauly to exploit market inefficiencies while avoiding outcome exposure—executing a fully hedged strategy with minimized risk.

// Impact: Over three months, Pauly achieved a 13.79% PNL (~67.65% APY), exiting all positions before market resolution. This demonstrated the effectiveness of Allora-powered intelligence in driving safe, performant AI agents in DeFi.



Allora supports a broad range of applications by enabling model creators to optimize for virtually any target objective.


Allora